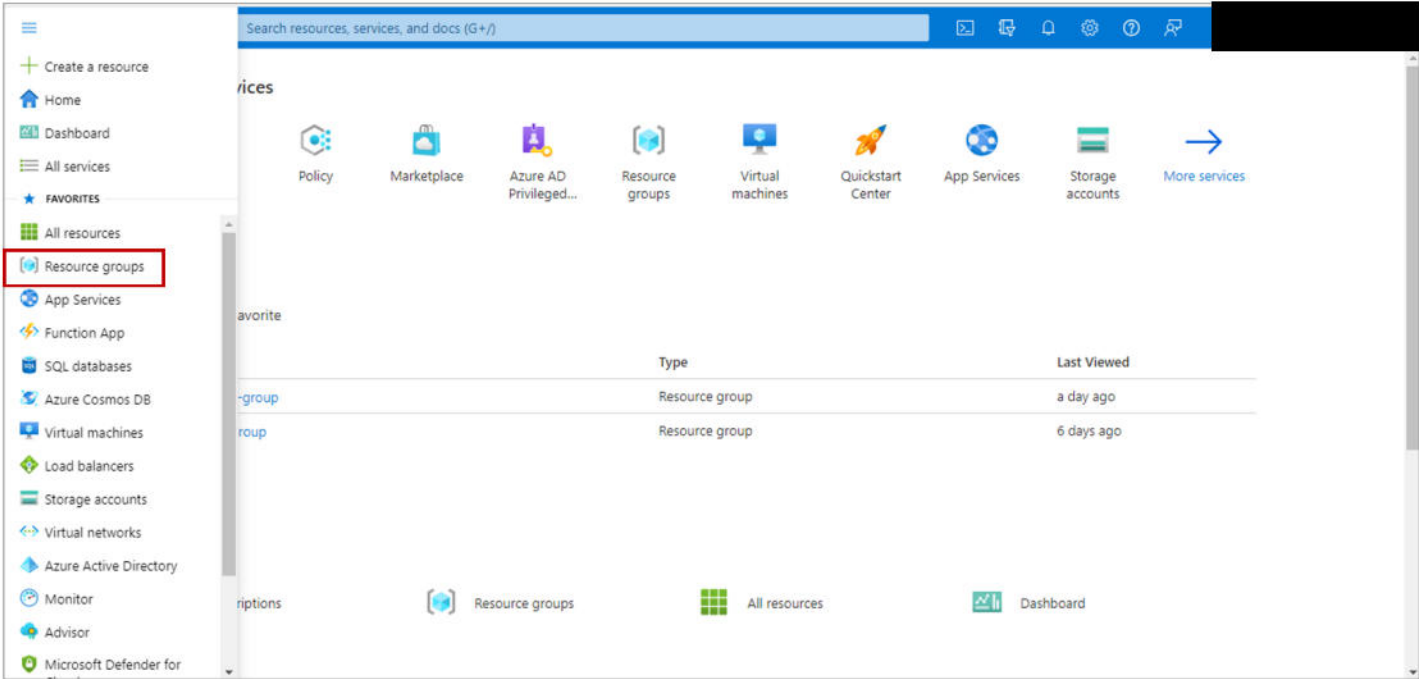


Audit logs project solution

Instructions: List the steps you will perform to complete each activity in the system. Provide screenshots for each step. As a sample, we have filled in the information for Activity 1, Step 1. You can add more rows as required.

Task: Configure audit logs.

Activity #	Steps						
Activity 1: Create a resource group.	1. Sign in to the Azure portal with your login credentials. On the portal menu, select Resource groups.  <table><thead><tr><th>Type</th><th>Last Viewed</th></tr></thead><tbody><tr><td>Resource group</td><td>a day ago</td></tr><tr><td>Resource group</td><td>6 days ago</td></tr></tbody></table>	Type	Last Viewed	Resource group	a day ago	Resource group	6 days ago
Type	Last Viewed						
Resource group	a day ago						
Resource group	6 days ago						

Microsoft Azure

Upgrade

Search resources, services, and docs (G+/)

Copilot

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Home > Resource Manager

Resource Manager | Resource groups

Default Directory

Search

Resource Manager

All resources

Favorite resources

Recent resources

Resource groups

Tags

Organization

Tools

Deployments

Help

Summarize my costs by service

Export resource groups using Bicep or Terraform

+1

+

Create

Manage view

Refresh

Export to CSV

Open query

Assign tags

Group by none

Filter for any field...

Subscription equals all

Location equals all

Add filter

Name ↑	Subscription	Location
☐	AZ500LAB01	...
☐	Azure subscription 1	East US

Showing 1 - 1 of 1. Display count: auto

Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

Activity 2:
Create Azure Database for MySQL servers.

1. Flexible My SQL server has been selected

Microsoft Azure

Upgrade

Search resources, services, and docs (G+/)

Copilot

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Home > Azure Database for MySQL servers > Select Azure Database for MySQL deployment option

Flexible server

Microsoft

Basics

Tags

Review

Create an Azure Database for MySQL server.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ Azure subscription 1

Resource group * ⓘ AZ500LAB01

Create new

Server details

Enter required settings for this server, including picking a location and configuring the compute and storage resources.

Server name * ⓘ mysqlserverflexible1

Region * ⓘ (US) East US

Availability zone * ⓘ No preference

Authentication

Administrator login * ⓘ mysqlserver

Password * ⓘ

Estimated costs

Compute Sku

Standard_B1ms

USD 9.93/month

9.93

Storage

USD 2.30/month

Storage selected 20 GiB (USD 0.115 per GiB)

20 x 0.115

0.115

Auto scale IOPS

Auto scale IOPS is billed on usage in per million request increments. Learn more

Backup Retention

Backup retention is billed based on additional storage used for retaining backups. Learn more

Bandwidth

For outbound data transfer across services in different regions will incur additional charges. Any inbound data transfer is free. Learn more

Estimated total

USD 12.23/month

Review + create

Next : Tags >

2. Review+create page of the My SQL server

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot

Home > Azure Database for MySQL servers > Select Azure Database for MySQL deployment option

Flexible server

Microsoft

Basics

Subscription	Azure subscription 1
Resource group	AZ500LAB01
Server name	mysqlserverflexible1
Administrator login	mysqlserver
Location	East US
Availability zone	No preference
High availability	Disabled
MySQL version	8.4
Workload type	Dev/Test
Compute + storage	Burstable, 8 vCores, 2 GiB RAM, 20 GiB storage, Auto scale IOPS
Backup redundancy	Locally-redundant
Zonal Resiliency	No

Additional configuration

Server parameter - lower_case_table_names	1
---	---

Tags

ⓘ If you need to modify the default settings, please click on [Advanced Create](#)

Create < Previous : Tags [Download a template for automation](#)

Auto scale IOPS

Auto scale IOPS is billed on usage in per million request increments. [Learn more](#)

Backup Retention

Backup retention is billed based on additional storage used for retaining backups. [Learn more](#)

Bandwidth

For outbound data transfer across services in different regions will incur additional charges. Any inbound data transfer is free. [Learn more](#)

Estimated total **USD 12.23/month**

Prices reflects an estimates only. [View Azure pricing calculator](#). Final charges will appear in your local currency in cost analysis and billing views.

3. Deployment of the server is **successful**.

The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the 'Microsoft Azure' logo, an 'Upgrade' button, a search bar, and a 'Copilot' button. The left sidebar contains a navigation menu with options like 'Create a resource', 'Home', 'Dashboard', 'All services', and a 'FAVORITES' section listing various Azure services. The main content area is titled 'MySQLFlexibleServer_89268130750e11f1bc111d66e04e23f5 | Overview'. It features a search bar and action buttons: 'Delete', 'Cancel', 'Redeploy', 'Download', and 'Refresh'. The 'Overview' tab is selected, showing a green checkmark and the message 'Your deployment is complete'. Below this, deployment details are listed: 'Deployment name : MySQLFlexibleServer_89268...', 'Start time : 7/1/2026, 10:34:38 AM', 'Subscription : Azure subscription 1', and 'Correlation ID : f1de053b-f136-4900-a28f-7...'. There are also links for 'Deployment details' and 'Next steps'. On the right, there are two informational panels: 'Cost management' with a link to 'Set up cost alerts >', and 'Azure Database Migration Service' with links to 'Get started with MySQL Flexible Server', 'Minimal downtime migration into Azure Database for MySQL with Azure DMS', and 'Free Microsoft tutorials'.

Microsoft Azure Upgrade Search resources, services, and docs (G+/I) Copilot

Home

MySQLFlexibleServer_89268130750e11f1bc111d66e04e23f5 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : MySQLFlexibleServer_89268... Start time : 7/1/2026, 10:34:38 AM

Subscription : Azure subscription 1 Correlation ID : f1de053b-f136-4900-a28f-7...

Resource group : AZ500LAB01

> Deployment details

✓ Next steps

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

[Set up cost alerts >](#)

Get started with MySQL Flexible Server

Azure Database Migration Service

Azure Database Migration Service is a fully-managed Azure service that helps you easily migrate various databases to their corresponding Azure data services.

[Minimal downtime migration into Azure Database for MySQL with Azure DMS](#)

Free Microsoft tutorials

[Create PHP + MySQL App](#)

[Migrate RDS MySQL to Azure Database for MySQL](#)

[Physical Migration](#)

[Azure Database for MySQL Import CLI](#)

Add or remove favorites by pressing Ctrl+Shift+F

Activity 3:
Create a Key Vault. Then, create a user-assigned managed identity and set permissions for key operations.

1. Creating a key and giving a user **Get, List, Wrap, and Unwrap** permissions only.

Grant data plane access by using a [Azure RBAC](#) or [Key Vault access policy](#)

☐

Azure role-based access control (recommended) ⓘ

☒

Vault access policy ⓘ

Resource access

☐

Azure Virtual Machines for deployment ⓘ

☐

Azure Resource Manager for template deployment ⓘ

☐

Azure Disk Encryption for volume encryption ⓘ

Access policies

Access policies enable you to have fine grained control over access to vault items. [Learn more](#)

+ Create

✎ Edit

🗑 Delete

☒

Name ↑↓

☐

Email ↑↓

☐

Key Permissions

☐

Secret Permissions

☐

Certificate Permissions

▼ USER

☒

Charlie Chaplin

Get, List, Update, Create, Import, Del...

Get, List, Set, Delete, Recover, Backu...

Get, List, Update, Create, Import, Del...

Previous

Next

Review + create

[Give feedback](#)

2. Created a user-assigned managed identity to use later on in the My SQL server

The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the 'Microsoft Azure' logo, an 'Upgrade' button, a search bar, and a 'Copilot' button. The left sidebar contains a 'Create a resource' button and a list of services under 'FAVORITES', including Application Templates, All resources, Resource groups, App Services, Function App, Azure SQL Database, Azure Cosmos DB, Virtual machines, Load balancers, Storage accounts, Virtual networks, Microsoft Entra ID, Monitor, Advisor, Microsoft Defender for Cloud, and Cost Management + Billing.

The main content area shows the 'Overview' page for a managed identity named 'demo-umi' in the 'AZ500LAB01' resource group. The page includes a search bar, a 'Delete' button, and a 'JSON View' link. The 'Overview' tab is selected, displaying a list of settings and their values:

Property	Value
Resource group	AZ500LAB01
Location	East US
Subscription	Azure subscription 1
Subscription ID	ae057139-3ee2-4662-a3fb-11c8432a3437
Type	Microsoft.ManagedIdentity/userAssignedIdentities

Below the settings, there is a section for 'Essentials' with the following details:

Property	Value
Client ID	37609b53-3102-4a22-b79f-5f6c7109b489
Object (principal) ID	68717962-ca89-464b-ab33-69b351d1b878
Isolation Scope	None
Resource	None

At the bottom of the page, there is a note: 'Add or remove favorites by pressing Ctrl+Shift+F.'

3. Created access policy in the key vault for demo-umi

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot

Home > AZ500LAB01 > Demo-Keyvault10

Demo-Keyvault10 | Access policies

Key vault

Search Create Refresh Delete Edit

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Access policies
Resource visualizer
Events
Objects
Keys
Secrets
Certificates
Settings
Monitoring
Automation
Help

Access policies enable you to have fine grained control over access to vault items. [Learn more](#)

Search Permissions: All Type: All

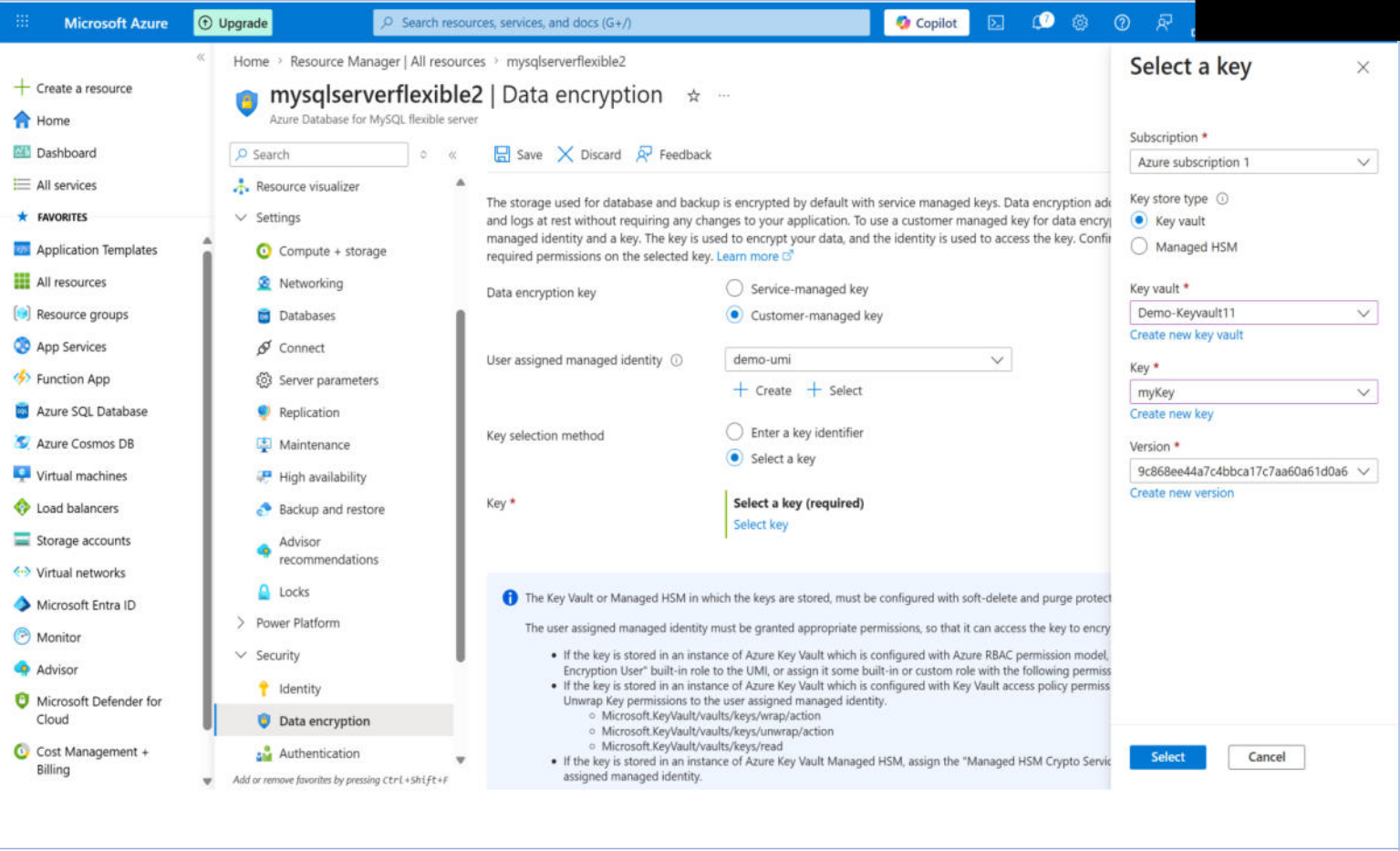
Showing 1 to 2 of 2 records.

☐	Name ↑	Email ↑	Key Permissions	Secret Permissions	Certificate Permissions
☐	APPLICATION				
☐	demo-umi		Unwrap Key, Wrap Key, Get,...		
☐	USER				
☐	Charlie Chaplin		Get, List, Wrap Key, Unwrap...		

Add or remove favorites by pressing Ctrl+Shift+F

Activity 4:
Configure customer-managed key.

1. Assigned the managed identity to the SQL Server for Azure Key Vault access, it shows the **Data encryption** page for the flexible server with the Key Vault and key names entered in the **Select a key** pane.

	 The screenshot displays the Azure portal interface for configuring data encryption on a MySQL flexible server. The left sidebar shows the navigation menu with 'Data encryption' selected under the 'Security' category. The main content area shows the 'Data encryption' settings for the resource 'mysqlserverflexible2'. The 'Data encryption key' is set to 'Customer-managed key', and the 'User assigned managed identity' is 'demo-umi'. A 'Select a key' dialog is open on the right, allowing the user to choose a key from the 'Demo-Keyvault11'. The dialog shows the key 'myKey' and its version '9c868ee44a7c4bbca17c7aa60a61d0a6'. A blue 'Select' button is visible at the bottom of the dialog. An information box at the bottom of the settings page provides details on key permissions and soft-delete/purge protection.
Activity 5: Use data encryption for restore.	1. Create Azure Database for MySQL Flexible server – Restore server page with a name and Compute + storage details entered for the restore server

Basics Networking Security Tags Review + create

Create an Azure Database for MySQL flexible server from an existing backup. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription ^① Azure subscription 1 ▼

Resource group ^① AZ500LAB01 ▼

[Create new](#)

Source details

Select a backup source and detail. Additional settings will be defaulted where possible based on the backup selected.

Source server ^① mysqlserverflexible2

Geo-redundant restore ^① ☐ Restore to cross region

i Geo-redundant backup is not configured for the source server (mysqlserverflexible2)

Earliest restore point ^① 2026-07-01 05:45:02.1803836 UTC

Point-in-time-restore (PITR) ^① ☒ Latest restore point (Now)

☐ Select a custom restore point

☐ Select fastest restore point (Restore using full backup)

Server details

Enter required settings for this server, including picking a location and configuring the compute and storage resources.

Name * ^① restoremysqlserverflexible2 ✓

Location ^① East Asia ▼

MySQL version ^① 8.4 ▼

Availability zone ^① 1 ▼

Compute + storage ^① **Burstable, B1ms**
1 vCores, 2 GiB RAM, 20 GiB storage
[Configure server](#)

Estimated costs



^ Compute	USD 20.88/month
Free upto 750 hours	
Standard_B1ms (1 vCore)	20.88
^ Storage	USD 3.00/month
Free upto 32 GB	
20 GiB (USD 0.15 per GiB)	20 x 0.15
^ Auto scale IOPS	
Auto scale IOPS is billed on usage in per million request increments. Learn more	
^ Backup Retention	
Backup retention is billed based on additional storage used for retaining backups. Learn more	
^ Bandwidth	
Outbound data transfer across services in different regions will incur additional charges. Any inbound data transfer is free. Learn more	

Estimated total **USD 23.88/month**

Charges will apply if you use above the free monthly limits. Please check [your usage](#) of free services. Final charges will appear in your local currency.

2. Same customer-managed key assigned to the restore server of our original My SQL server

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot

Home > restoremysqlserverflexible2

restoremysqlserverflexible2 | Data encryption

Azure Database for MySQL flexible server

Search Save Discard Feedback

Connect

- Server parameters
- Replication
- Maintenance
- High availability
- Backup and restore
- Advisor recommendations
- Locks

> Power Platform

▼ Security

- Identity
- Data encryption**
- Authentication
- Microsoft Defender for Cloud

> Monitoring

> Automation

> Help

Add or remove favorites by pressing Ctrl+Shift+F

The storage used for database and backup is encrypted by default with service managed keys. Data encryption additionally encrypts your databases, backups, and logs at rest without requiring any changes to your application. To use a customer managed key for data encryption, you must select a user assigned managed identity and a key. The key is used to encrypt your data, and the identity is used to access the key. Confirm that the identity selected is granted the required permissions on the selected key. [Learn more](#)

Data encryption key

☐ Service-managed key

☒ Customer-managed key

User assigned managed identity

demo-umi

+ Create + Select

Key selection method

☐ Enter a key identifier

☒ Select a key

Key *

myKey/139fdb995bf24c808358618d3495e5b4

<https://demo-keyvault11.vault.azure.net/keys/myKey/139fdb995bf24c808358618d3495e5b4>

[Select key](#)

The Key Vault or Managed HSM in which the keys are stored, must be configured with soft-delete and purge protection features enabled.

The user assigned managed identity must be granted appropriate permissions, so that it can access the key to encrypt and decrypt the data.

- If the key is stored in an instance of Azure Key Vault which is configured with Azure RBAC permission model, assign the "Key Vault Crypto Service Encryption User" built-in role to the UMI, or assign it some built-in or custom role with the following permissions:
- If the key is stored in an instance of Azure Key Vault which is configured with Key Vault access policy permission model, grant the Get, List, Wrap Key, and Unwrap Key permissions to the user assigned managed identity.
 - Microsoft.KeyVault/vaults/keys/wrap/action
 - Microsoft.KeyVault/vaults/keys/unwrap/action
 - Microsoft.KeyVault/vaults/keys/read
- If the key is stored in an instance of Azure Key Vault Managed HSM, assign the "Managed HSM Crypto Service Encryption User" built-in role to the user

Activity 6:

Configure an audit log for connection and general events.

1. Enabling “configure_audit_logs” in the server parameters so that we can record the database activities.

The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the 'Microsoft Azure' logo, an 'Upgrade' button, a search bar, and a 'Copilot' button. The left sidebar contains a list of services and favorites, including 'Create a resource', 'Home', 'Dashboard', 'All services', 'FAVORITES', 'Application Templates', 'All resources', 'Resource groups', 'App Services', 'Function App', 'Azure SQL Database', 'Azure Cosmos DB', 'Virtual machines', 'Load balancers', 'Storage accounts', 'Virtual networks', 'Microsoft Entra ID', 'Monitor', 'Advisor', 'Microsoft Defender for Cloud', and 'Cost Management + Billing'. The main content area shows the 'Overview' page for a deployment named 'MySQLFlexibleServerParameters_5369552a7a3549d7a0610de642072d0d'. The deployment status is 'Your deployment is complete'. The deployment details show the name, subscription, resource group, start time, and correlation ID. The 'Next steps' section provides links to discover how to migrate MySQL databases, learn how to setup Azure Database Migration Service, learn how to manage the server, setup a firewall rule, learn about private access connectivity, and setup monitoring alerts. The right sidebar contains sections for 'Cost management', 'Get started with MySQL Flexible Server', 'Azure Database Migration Service', and 'Free Microsoft tutorials'.

2. I then enabled Connection and General in the “audit_log_events” parameter to specify what kind of activity do I want to record in the My SQL server.

Microsoft Azure

Upgrade

Search resources, services, and docs (G+/J)

Copilot

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

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Tags

Diagnose and solve problems

Learning center

Resource visualizer

Settings

Compute + storage

Networking

Databases

Connect

Server parameters

Replication

Maintenance

High availability

Backup and restore

Advisor recommendations

Home > MySQLFlexibleServerParameters_5369552a7a3549d7a0610de642072d0d | Overview > mysqlserverflexible2

mysqlserverflexible2 | Server parameters

Azure Database for MySQL flexible server

Search

Save Discard Reset all to default Feedback

There are 1 unsaved parameter changes. Please save to apply these updates.

Top Modified Read-Only All

Top tab displays commonly modified parameters. You can hover over the info icon to get details about the allowed values of a particular parameter. For a comprehensive list of all available parameters, navigate to All tab [Learn more](#)

Search to filter items in Top Tab...

Parameter name	Value	Parameter type	Description
audit_log_enabled		Dynamic	Allow to audit the log.
audit_log_events	CONNECTION	Dynamic	Select the events to audit logs.
binlog_expire_logs_seconds		Dynamic	Number of seconds to service waits before the binary log file get...
binlog_row_image		Dynamic	For MySQL row-based replication, this variable determines how r...
character_set_server		Dynamic	Use charset_name as the default server character set.
collation_server		Dynamic	The server's default collation.
connect_timeout		Dynamic	The number of seconds that the mysqld server waits for a connec...
enforce_gtid_consistency		Dynamic	When enable, this option enforces GTID consistency by allowing ...
gtid_mode		Dynamic	Indicates if global transaction identifiers (GTIDs) are used to iden...
init_connect		Dynamic	A string to be executed by the server for each client that connects.
innodb_buffer_pool_size		Dynamic	The size in bytes of the buffer pool, the memory area where Inno...
innodb_io_capacity	10000	Dynamic	Sets an upper limit on I/O activity performed by InnoDB backgro...

Add or remove favorites by pressing Ctrl+Shift+F

3. The parameter was successfully saved.

The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the 'Microsoft Azure' logo, an 'Upgrade' button, a search bar, and a 'Copilot' icon. The left sidebar lists various services and favorites, including 'Create a resource', 'Home', 'Dashboard', 'All services', and a 'FAVORITES' section with items like 'Application Templates', 'All resources', 'Resource groups', 'App Services', 'Function App', 'Azure SQL Database', 'Azure Cosmos DB', 'Virtual machines', 'Load balancers', 'Storage accounts', 'Virtual networks', 'Microsoft Entra ID', 'Monitor', 'Advisor', 'Microsoft Defender for Cloud', and 'Cost Management + Billing'.

The main content area shows the 'Overview' page for a deployment named 'MySQLFlexibleServerParameters_c6cd9c22ed644e52883d884fe4cf64b1'. The deployment status is 'Complete', indicated by a green checkmark. The deployment details include the name, start time (7/1/2026, 11:56:20 AM), subscription (Azure subscription 1), correlation ID, and resource group (AZ500LAB01). The 'Next steps' section provides links to learn more about migrating MySQL databases, setting up Azure Database Migration Service, managing the server, and setting up monitoring alerts. The right sidebar features a 'Cost management' section with a budget alert icon and a 'Get started with MySQL Flexible Server' link, followed by a section for 'Azure Database Migration Service' and 'Free Microsoft tutorials'.

Deployment Details:

- Deployment name: MySQLFlexibleServerParam...
- Start time: 7/1/2026, 11:56:20 AM
- Subscription: Azure subscription 1
- Correlation ID: 5d2b795a-4782-40ce-ad41-...
- Resource group: AZ500LAB01

Next steps:

- Discover how to migrate MySQL databases with Azure Database Migration Service Recommended
- Learn how to setup Azure Database Migration Service Recommended
- Learn how to manage your server Recommended
- For public access connectivity, setup a firewall rule to connect to the server Recommended
- Learn about private access connectivity method Recommended
- Setup monitoring alerts Recommended

Cost management:

Get notified to stay within your budget and prevent unexpected charges on your bill. [Set up cost alerts >](#)

Get started with MySQL Flexible Server

Azure Database Migration Service

Azure Database Migration Service is a fully-managed Azure service that helps you easily migrate various databases to their corresponding Azure data services. [Minimal downtime migration into Azure Database for MySQL with Azure DMS](#)

Free Microsoft tutorials

- [Create PHP + MySQL App](#)
- [Migrate RDS MySQL to Azure Database for MySQL](#)
- [Physical Migration](#)
- [Azure Database for MySQL: Import CLI](#)